Cement Board Installation Guide

Interior Application

SECO LITE®



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SECOLITE®

The Future of Cement Board



SECOLITE® Cement Board is a light-weight board made of Portland cement, aggregates and other inorganic materials, reinforced on both sides with embedded glass fiber.

SECOLITE® Cement Board is ideal for use in demanding environments. It can be used for both indoor and outdoor applications.

Water resistant
Weatherproof for up to 12 months

Impact resistant
Mould resistant

Non-combustible
according to EN13501-1
Excellent fire performance

Can be 'Score and Snapped'
Dimensionally stable





SECOLITE® Cement Board is designed to be score and snapped as opposed to traditional cement boards that require power tools and dust extraction to cut. The make-up of the board improves the usability when compared to other, traditional boards that are heavier, harder to handle and can generate large amounts of dust when cut.

SECOLITE® Cement Board is produced using a unique manufacturing process.

SECOLITE® Cement Board is used in the construction of interior and exterior walls, ceilings and floors.





Installation

Easy, light and robust. SECOLITE® Cement Board is a tough, impact-resistant glass fiber reinforced cement board for interior use in wet and humid areas.

Internal areas that are exposed to moisture and high humidity are particularly prone to damage from water. Where materials such as plasterboard or plywood can break down or warp when exposed to prolonged periods of moisture SECOLITE® Cement Board remains dimensionally stable and maintains performances. Saving cost and issues in the long term.

SECOLITE® is the perfect board for use even in the most challenging wet and humid conditions for both wall and ceiling applications, making it ideal for everything from swimming pools and saunas to communal showers and kitchen areas.

Manufactured from inorganic materials and highly resistant to water and mould, SECOLITE® Cement Board weighs just 13.6 kg/m² per square meter – making it easy to handle and fast to install.





Installation Dry walls

Substructure and board alignment

Ensure metal or timber framing is suitably fixed to the surrounding structure. For best results, use lightweight SECOLITE® metal studs. Metal or timber profiles should be fixed at no more than 600mm centres. Additional horizontal support or reduced stud centres may be required depending on specific performance requirements.

Align the first SECOLITE® Cement Board panel along the framing profiles using a level. Ensure that board joints align with the centre of the studs.



Horizontal installation of SECOLITE® Cement Boards

Cutting

SECOLITE® Cement Board can be cut using the 'score & snap' method. There is no need for power tools and dust extraction.

Once the desired dimensions have been marked on the SECOLITE® Cement Board using a sharp knife, score the board to one side ensuring that the mesh is cut. Bend and snap along the initial cut, then use the knife to slice through the mesh at the back.



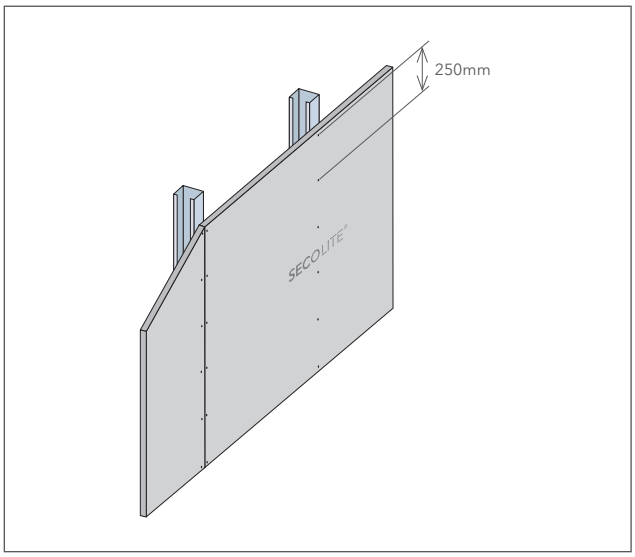
Snap and cut the mesh through on the reverse

Fixing

Once the SECOLITE® Cement Board has been cut to size if required, offer up the board in the vertical or horizontal position with framing members at maximum 600 mm centres. Ensure that the SECOLITE® Cement Board is located on the centre line of the framing member.

Once the board is located, use SECOLITE® Screws to fix the board to the framing members. SECOLITE® Cement Boards should be fixed at a maximum of 250mm centres and 15mm in from all board edges.

Screws should not be over tightened and sit flush with the surface.



SECOLITE® Cement Board Layout

Jointing and skim-coating

Suitable for making the joints of SECOLITE® Cement Boards are joint fillers and skimming plasters on gypsum- or acrylic basis as well as ready mix products (ready to use in pails) on calcium carbonate basis. These products can also be used for the skimming of the full board surface.

Push the filler between the gaps in the boards and spread it across the face of the board. Firmly embed SECOLITE® Joint Tape into the centre of the joints (both vertical and horizontal joints). When the SECOLITE® Joint Tape is secured scrape away excess jointing material leaving the SECOLITE® Joint Tape covered. Spot the screw heads with the filler. Allow the filler to set as per the application guidelines of the individual product.



SECOLITE® Joint Adhesive

Once the joint is dried, apply a 2-3 mm layer of skim-coat to the entire surface of the board / wall using one of the following materials:

- Acrylic joint fillers and skimming plasters
- Ready-mix products (ready to use in pails) on calcium-carbonate basis.
- Gypsum based joint fillers, skimming plasters or thin coats (not recommended for wet areas)



Embedment of Joint Tape



Skim-coating

Smooth out the full surface and once dry, the surface can be painted.

Most common painting systems can be used. These include water-based emulsions, matt enamels, polymer resin paints, and epoxy-based enamels.

Jointing interior walls with tile finishes

Apply SECOLITE® Joint Adhesive on the edge of the first board that is fixed on the studs and push the following board firmly against the first board. Hold the second board in place and fasten it with SECOLITE® Screws at a maximum of 250mm centres and 15mm in from all board edges.

Allow the joint adhesive to dry and scrap away excess material.

Use a flexible tile adhesive for applying ceramic tiles (size ≤ 600 mm x 600 mm). SECOLITE® Cement Boards should be primed before applying the tile adhesive. The tile adhesive must meet the requirements of class C2 according to EN 12004. Different instructions apply for the installation of heavy tiles >50 kg/m².

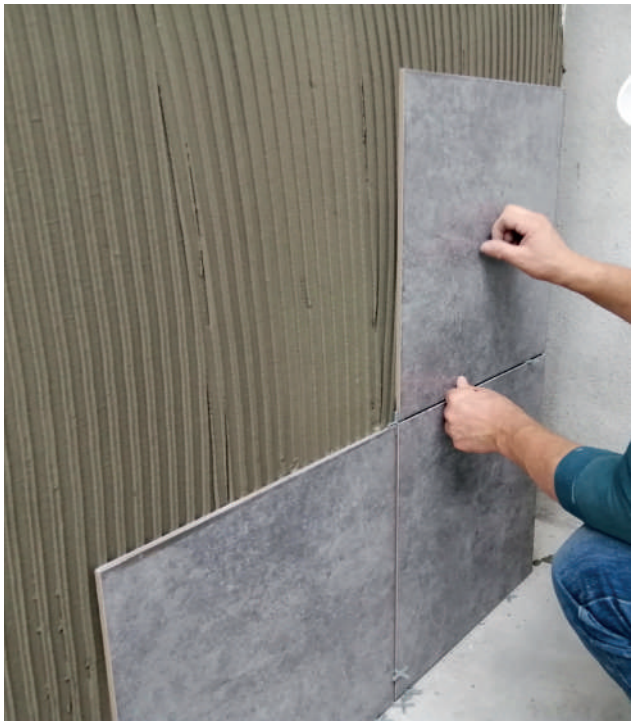
Vapour barriers in wet rooms

We recommend to install a vapour barrier membrane on the backside of the board around the perimeter of the room.

The type of membrane depends on the requirements as specified by the architect.



Scrape away excess joint adhesive



Tile application on SECOLITE® Cement Board

Installation
Floor Tile Underlay

Traditional plywood methods of tile backing can warp and cause tile failure. The installation of SECOLITE® Cement Board is an effective way to avoid severe damage from tile failure.

In case of tile or grout failure, only simple localised repair is needed as water ingress will not harm SECOLITE® Cement Board as an underlay.

Create the substructure

SECOLITE® Cement Board is installed on top of a sub-floor made from OSB or Chipboard that is fixed on a substructure from timber battens or steel profiles.

The minimum sub-floor thickness depends on the centre distance of the steel /timber joists.

Minimum subfloor thickness

Profile centers	400 mm	600 mm
OSB	18 mm	18 mm
Chipboard	18 mm	22 mm

Apply adhesive and lay the boards

SECOLITE® Cement Boards with 12.5 mm or 8 mm thickness can be used. SECOLITE® Handy Boards (e.g. 1200 x 800 mm) are the preferred option due to low weight and ease of handling.

Apply a complete bed of a suitable tile adhesive to the sub-floor.

The first board needs to be fitted from the point farthest away from the door.

The boards need to be laid in the following way:

- Joints are staggered at minimum 200 mm between the SECOLITE® Cement Board and the sub-floor.
- No two joints should coincide.
- A gap of 3 mm is left between each board.

Fixing and jointing

Use SECOLITE® Cement Board Screws SN 4 x 32 at a maximum of 250mm centres and 15mm in from all board edges. Fill the joints with SECOLITE® Joint Adhesive and let it dry. Scrape excess material away afterwards. Alternatively, the joints can be filled with a suitable tile adhesive.

Tiling

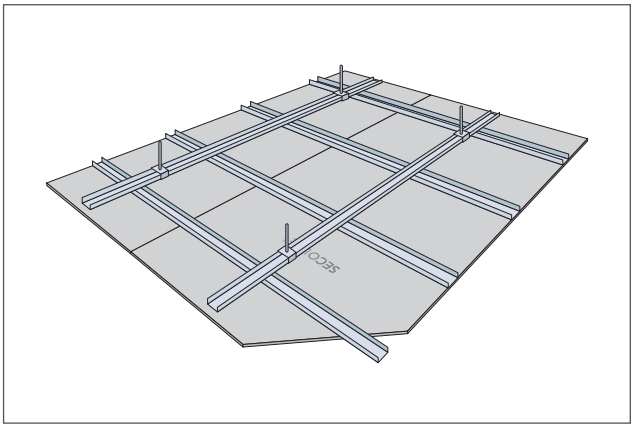
Apply a suitable tile adhesive to the board surface and position the tiles.

Installation Suspended Ceilings

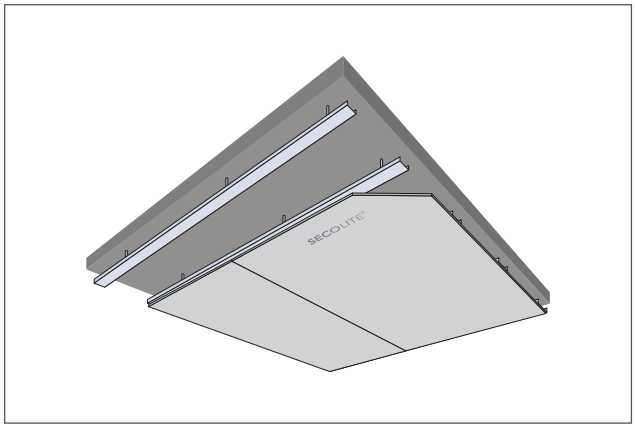
The installation of SECOLITE® Cement Boards in suspended ceilings follows the same rules as for walls.

To ensure optimum results, we recommend to use of SECOLITE® Metal Profiles at a maximum spacing of 400 mm.

SECOLITE® Cement Boards should be fixed at a maximum of 200 mm centres. Screws should be fixed 15 mm from the edge of the board and set flush with the board face without over driving.



Isometric showing a ceiling installation using SECOLITE® Cement Boards



Plan view of a ceiling installation using SECOLITE® Cement Board

Secolite® Cement Boards Handling

SECOLITE® Cement Boards are delivered on pallets up to 56, 40 and 30 for product thicknesses of 8, 12.5 and 15 mm. A forklift capable to safely lift 2,000 kg should be used to lift and move pallets of boards. Ensure that storage areas can support the weight of the pallets and boards (up to 1,500 kg/pallet). Do not stack SECOLITE® Cement Board more than 6 pallets high.

SECOLITE® should be stored in a dry location and protected from prolonged exposure to excess moisture. Allow boards to acclimate to ambient moisture and humidity conditions prior to installation.



Secolite® Cement Boards

SCL 8



Dimensions*	Tech. Data		Trucks 24 tons load			Containers 40/45'HCPW - 26 tons load		
	kg/m²	m²/ board	boards/ pal.	boards/ truck	m²/ truck	boards/ pal.	boards	m²/ cont.
8 x 2 000 x 1 200	9.2	2.40	56	1 064	2 554	56	1 120	2 688
8 x 2 400 x 1 200	9.2	2.88	56	896	2 580	56	952	2 742

** On request, other dimensions available / possible*

SCL 12.5



Dimensions*	Tech. Data		Trucks 24 tons load			Containers 40/45'HCPW - 26 tons load		
	kg/m²	m²/ board	boards/ pal.	boards/ truck	m²/ truck	boards/ pal.	boards	m²/ cont.
12.5 x 2 000 x 1 200	13.6	2.40	40	720	1 728	40	760	1 824
12.5 x 2 400 x 1 200	13.6	2.88	36	576	1 659	37	629	1 812
12.5 x 2 500 x 1 200	13.6	3.00	35	560	1 680	36	612	1 836

** On request, other dimensions available / possible*

SCL 15



Dimensions*	Tech. Data		Trucks 24 tons load			Containers 40/45'HCPW - 26 tons load		
	kg/m²	m²/ board	boards/ pal.	boards/ truck	m²/ truck	boards/ pal.	boards	m²/ cont.
15 x 2 000 x 1 200	16.2	2.40	30	600	1 440	30	648	1 555
15 x 2 400 x 1 200	16.2	2.88	30	480	1 382	30	540	1 555

** On request, other dimensions available / possible*

SCL 12.5 HANDYBOARDS

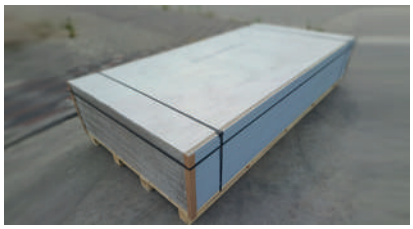


Dimensions*	Tech. Data		Trucks 24 tons load			Containers 40/45'HCPW - 26 tons load		
	kg/m²	m²/ board	boards/ pal.	boards/ truck	m²/ truck	boards/ pal.	boards	m²/ cont.
12.5 x 1 200 x 800	13.6	0.96	90	1 800	1 728	90	1 890	1 814
12.5 x 1 000 x 900**	13.6	0.90	90	1 890	1 701	90	2 070	1 863
12.5 x 1 000 x 800**	13.6	0.80	90	2 160	1 728	90	2 250	1 800
12.5 x 1 000 x 1 800**	13.6	1.80	60	960	1 728	60	1 020	1 836
12.5 x 900 x 1 800**	13.6	1.62	66	1 056	1 711	66	1 122	1 818
12.5 x 1 000 x 1 500**	13.6	1.50	60	1 140	1 710	60	1 200	1 800

** On request, other dimensions available / possible*
*** All four edges are cut*

Availability and lead time: product is ready for shipping 4-6 weeks after receiving the order confirmation for orders up to 5 trucks/containers.

Lead time for orders > 5 trucks/ containers on request.



SECOLITE® Pallet



SECOLITE® HANDYBOARDS Pallet

Secolite® Complementary Products

SCL Cement Board Screws

** Net usage w/o waste, spare and extras*



Description	Dimensions	Packing	Exterior	Interior	Application	Unit	Usage
PN. Designed for fixing SECOLITE® Cement Board on timber and metal frame works (thickness up to 0.7 mm). Min. screw-in depth: 10 mm for metal and 25 mm for timber applications.	4x32 / 41	1000 pcs / box	✓	✓	Wall: profile spacing 600 / 625	pcs/m²	15*
					Ceiling: profile spacing 300 / 312.5		25*
SD. Designed for fixing SECOLITE® Cement Board on metal frame works with thicknesses 0.8 to 2.0 mm.	4x32 / 41	1000 pcs / box	✓	✓	Wall: profile spacing 600 / 625	pcs/m²	15*

SCL Joint Tape



Description	Dimensions	Packing	Exterior	Interior	Application	Unit	Usage
Joint mesh tape for reinforcement of exterior and interior joints made with SECOLITE® Joint Filler & Skim Coat.	125 mm wide	1 500 m² / pallet; 240 rolls / pallet	✓	✓	Wall / Ceiling	lm / m²	1.5

SCL Joint Filler And Skim Coat



Description	Dimensions	Packing	Exterior	Interior	Application	Unit	Usage
Cement based filling material for joints and full-surface skimcoating, interior & exterior application.	Bags 20 kg	60 bags / pallet	✓	✓	Joint	kg / m²	0.6
				✓	Ceiling / Wall: full-surface filling	kg / m² / mm	0.6
						kg / m²	1.8
			✓		Wall / Ceiling: full-surface filling 2.5-3.5 mm incl. Inlay of reinforcement mesh	kg / m² / mm	1.6
						kg / m²	4.8

SCL Joint Adhesive



Description	Dimensions	Packing	Exterior	Interior	Application	Unit	Usage
For the bonding of SECOLITE® Cement Board in the wall. Not used in the ceiling area.	310 ml/ cartridge			✓	Joint	ml/m joint	25
						m² of cement board 1 200 x 2 400 w/ 1 cartridge	11

SCL Water Resistive Barrier Membrane



Description	Dimensions	Packing	Exterior	Interior	Application	Unit	Usage
Light weight but strong breather membrane for exterior wall applications; excellent water resistance but still high breathability;	150 g/m²; Width:1.5m; Roll: 75m²	Roll	✓		Wall	m²/m²	1.25

SCL Reinforcement Mesh



Description	Dimensions	Packing	Exterior	Interior	Application	Unit	Usage
Coating, applied on Basecoat or Joint Filler & Skim Coat	Width: 1000 mm; 160 g/m²; 50 m²/roll	Roll	✓		Wall / Ceiling	m²/m²	1.19

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